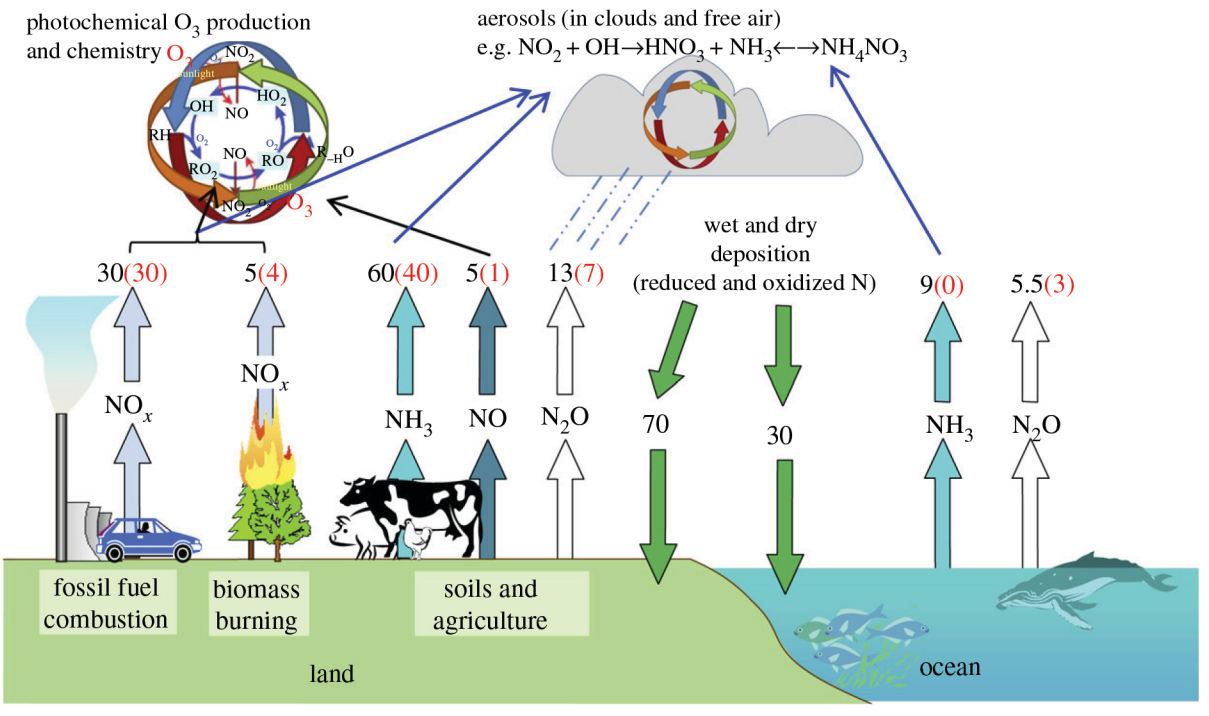
photochemical O_3 production and chemistry
O_3
NO_2
sunlight
OH
NO
HO_2
RH
O_2
NO_2
O_2
RO
R–H$_2$O
RO_2
sunlight
NO_2
O_2
O_3
aerosols (in clouds and free air)
e.g. $NO_2 + OH \rightarrow HNO_3 + NH_3 \longleftrightarrow NH_4NO_3$
wet and dry deposition (reduced and oxidized N)
30(30)
5(4)
60(40)
5(1)
13(7)
9(0)
5.5(3)
NO_x
NO_x
NH_3
NO
N_2O
70
30
NH_3
N_2O
fossil fuel combustion
biomass burning
soils and agriculture
land
ocean